

Results-Only Comparison of Models and Prompt Variants

Using `resource/results_variants.txt`

April 1, 2026

How to Read This Report

This report is intentionally results-only. It is based on `resource/results_variants.txt` and focuses on what the logged outputs show under verifier-style criteria:

- waypoint-count correctness,
- local progress toward the goal,
- kinematic plausibility at a qualitative level,
- obstacle-aware behavior in the right-obstacle case,
- and consistency across prompt variants.

Because the logged responses do not include `selected_waypoint_index`, and some entries are truncated or malformed, the cells below use qualitative judgments instead of strict numeric verifier pass/fail labels.

Prompt Legend

1. 01: baseline full
2. 02: no reasoning, keep examples
3. 03: no reasoning, one example
4. 04: no reasoning, no examples
5. 05: reasoning, no examples
6. 06: concise with reasoning
7. 07: concise no reasoning
8. 08: no reasoning, center bias
9. 09: reasoning, two examples
10. 10: minimal waypoints only

Cell Labels

- **Strong:** five waypoints, locally sensible, clear forward progress, and good task alignment.
- **Good:** mostly sensible, minor issues, but still usable.
- **Mixed:** some progress or correct structure, but notable semantic or stability issues.
- **Weak:** poor local planning quality, wrong directional behavior, malformed output, or clear instability.
- **Fail:** severe structural issue such as missing waypoint count or obviously unusable behavior.

Overall Ranking

Model	Row 0	Row 4	Summary
ChatGPT Mini	Strong	Strong	Most consistent model in the log. Stable across all prompt styles with clean 5-waypoint outputs.
Qwen 14B	Good–Strong	Good–Strong	Best Qwen model in the file. Usually sensible, with some prompt brittleness and a few regressive variants.
Qwen 7B untuned	Mixed	Mixed	Some usable outputs, but high prompt sensitivity and several semantically poor trajectories.
Qwen 7B FT post-hoc reasoning	Mixed–Weak	Mixed	Can produce plausible plans, but very inconsistent and often non-local.
Qwen 7B FT no reasoning	Weak	Weak	Least reliable family: repeated points, malformed outputs, four-waypoint failures, and unstable motion.

Row 0: No-Obstacle / Clear-Path Results

Model	01	02	03	04	05	06	07	08	09
ChatGPT Mini	Strong: centered small-step rollout, clean schema, strong verifier-style progress.	Strong: same stable center-line behavior without reasoning field.	Good: mild diagonal bias but still local and forward.	Good: larger steps, but still straight and useful.	Mixed: geometry is fine, but reasoning text is internally inconsistent with the waypoints.	Mixed: aggressive long-horizon jumps; still forward, but less local.	Mixed: same long-horizon compression as 06, less local than ideal.	Strong: center-bias prompt behaves cleanly and predictably.	Strong: strong center-line with reasoning present.
Qwen 7B untuned	Good: small diagonal rollout, locally plausible, but reasoning is truncated.	Good: similar to baseline and structurally clean.	Weak: starts at the obstacle coordinate; semantically suspicious for a clear-path case.	Mixed: valid five points, but overly coarse global-style path.	Weak: waypoints hug current position and reasoning is badly inconsistent/truncated.	Mixed: diagonal progress is plausible but drifts away from the clear center-progress hint.	Mixed: forward progress exists, but steps become too coarse and global.	Good: same usable diagonal baseline pattern.	Good: geometry line, reasoning categorized.
Qwen 7B FT post-hoc	Weak: essentially jumps to goal region immediately; not a local rollout.	Fail: malformed JSON plus near-goal behavior; structurally unreliable.	Mixed: consistent medium-range diagonal rollout, but not clearly local.	Weak: starts on obstacle location and then jumps to goal.	Weak: creeps near obstacle coordinates; poor task alignment.	Mixed: obstacle-anchored start then diagonal escape; not ideal for clear path.	Weak: giant non-local jumps in y .	Weak: immediate jump to goal region with extra overshoot.	Mixed: goal again; than still no
Qwen 7B FT no reasoning	Weak: repeated identical points and truncated reasoning.	Weak: all waypoints collapse directly to the goal.	Fail: only four waypoints.	Strong: simple centered rollout, one of the few clean cases.	Weak: repeated near-static points and self-contradictory reasoning.	Good: local straight rollout with mild center bias.	Good: same stable local rollout without reasoning.	Weak: again collapses all waypoints to the goal.	Weak: near-st and t reason
Qwen 14B	Strong: clean, centered, local rollout.	Strong: same stable center-line behavior.	Good: diagonal but still locally plausible.	Good: larger forward spacing, still valid and useful.	Mixed: waypoints are fine, but reasoning text reports inconsistent progress.	Mixed: plausible diagonal rollout, but drifts more than needed in the clear case.	Mixed: very long forward jumps, less local than baseline.	Strong: same clean baseline response.	Good: to 7B rollout but t reason

Row 4: Obstacle on Right / Obstacle-Avoidance Results

Model	01	02	03	04	05	06	07	08	09
ChatGPT Mini	Strong: forward progress with mild left correction; very plausible.	Strong: same general behavior, slightly more spread in y .	Mixed: stays fixed in x and barely reacts to right-side caution.	Weak: moves further right, opposite of the left-clearer cue.	Weak: same rightward drift as 04, plus unreliable reasoning.	Good: preserves progress and avoids large jumps, though lateral response is modest.	Good: forward progress is strong, but still somewhat center-locked.	Mixed: center-bias keeps x fixed and under-reacts to obstacle cue.	Good: local though conservative in avoidance.
Qwen 7B untuned	Good: small leftward correction with local progress; reasoning truncated.	Good: similar to baseline and usable.	Weak: moves right instead of left, poor obstacle response.	Weak: keeps moving right and outward, not away from hazard.	Weak: similar wrong-direction behavior plus inconsistent reasoning.	Mixed: plausible local spacing, but drifts right instead of left.	Weak: large forward leaps without convincing obstacle-aware correction.	Good: small leftward correction again, better than 03–07.	Good: baseline strategy, but ing true.
Qwen 7B FT post-hoc	Weak: barely moves; almost static around current state.	Weak: jumps to goal region near $y \approx 29$, not local.	Mixed: local diagonal motion, but biased right rather than left.	Good: small local leftward correction, one of its better row-4 outputs.	Good: similar to 04, leftward and local.	Good: same controlled leftward arc with steady spacing.	Good: similar local leftward correction.	Weak: mostly local early points, then a huge jump to $y = 29$.	Strong: row-4 in this family; leftward smooth.
Qwen 7B FT no reasoning	Weak: repeated same point, little meaningful motion.	Mixed: valid five points, but drifts right rather than strongly left.	Fail: only four waypoints.	Mixed: local and valid, but too little reaction to the obstacle cue.	Weak: tiny motion and truncated reasoning; barely responsive.	Weak: reverses y direction and moves away from forward progress.	Weak: same regression in y , with unexpected z fields.	Fail: only four waypoints.	Mixed: forward but too reason; limited chance in.
Qwen 14B	Strong: mild but correct leftward avoidance with local progress.	Strong: similar to baseline and structurally clean.	Mixed: moves to the right, so obstacle response is not ideal.	Fail: decreases y , strongly regressive for the task.	Mixed: forward motion exists, but reasoning claims zero progress and the geometry is too center-locked.	Strong: strongest leftward avoidance in the log while preserving progress.	Strong: same as 06, slightly simpler and still very good.	Strong: balanced mild leftward correction, good compromise between progress and caution.	Strong: to baseline slightly leftward.

Best Prompt Variants by Model

Model	Best prompts	row-0	Best prompts	row-4	Comment
ChatGPT Mini	01, 02, 08, 09, 10		01, 02, 10		Very robust overall; minimal prompt surprisingly strong.
Qwen 7B untuned	01, 02, 08, 10		01, 02, 08, 10		Needs careful prompt choice; one-example and no-example variants are unreliable.
Qwen 7B FT post-hoc	03 marginally	only	04, 05, 06, 07, 09, 10		Better in obstacle case than clear-path case, but still unstable.
Qwen 7B FT no reasoning	04, 06, 07		none strong	clearly	Fine-tuning appears to have hurt stability more than helped.
Qwen 14B	01, 02, 08		01, 02, 06, 07, 08, 09, 10		Best open-model result in the file.

Takeaways

- ChatGPT Mini is the strongest result in the file because it stays clean and useful across all prompt styles.
- Qwen 14B is the best Qwen model and is especially good in the right-obstacle environment when given concise or lightly guided prompts.
- Untuned Qwen 7B is sometimes usable, but it is too prompt-sensitive to trust broadly.
- The two fine-tuned 7B models are not winning in this log. One is erratic and non-local; the other has repeated-point failures and malformed outputs.
- For obstacle avoidance specifically, Qwen 14B with prompts 06, 07, 08, 09, or 10 looks strongest among the open models recorded here.

Compile Command

```
cd /home/prachit/Desktop/vlm-conformal/px4_ws/src/llm_drone/resource
pdflatex -interaction=nonstopmode model_prompt_variant_comparison_report.tex
```